BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 10221

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. If 7:9::x:27 then .....

(A) x=21 (B) x=3

(C) x=7 (D) x=81

Q2. Sum of the roots of $2x^2-3x-4=0$ is .....

(A) 2/4 (B) 3/2

(C) 2/3 (D) −3/2

Q3. Sin 105° lies in ..... quadrant.

(A) first (B) second

(C) third (D) fourth

Q4. If $x^2-x-1=0$ then x = ...

(A) $(-1\pm\sqrt{5})/2$ (B) $-1\pm\sqrt{5}/2$

(C) $(1\pm\sqrt{5})/2$ (D) $1\pm\sqrt{5}/2$

Q5. If a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is ..... to the circle at that point.

(A) tangent (B) secant

(C) cotangent (D) none of these

Q6. If two arcs of a circle are congruent then the corresponding chords are .....

(A) unequal
(B) concurrent
(C) equal
(D) none of these

Q7. How many tangents can be drawn from a point outside the circle?

(A) 1
(B) 2
(C) 3
(D) 4

Q8. In the given set of data 5,7,7,5,3,7,2,8,2 mode is .....

(A) 9
(B) 5
(C) 2
(D) 7

Q9. The number of times a value appears on a set of data is called .....

(A) average
(B) mode
(C) frequency
(D) median

Q10. $1+\tan^2\theta$ = ...

(A) $\cos^2\theta$
(B) $\sin^2\theta$
(C) $\operatorname{cosec}^2\theta$
(D) $\sec^2\theta$

Q11. A chord 40 units long is contained in a circle of radius 25 units. The distance of the chord from the centre of the circle is ..... units.

(A) 15
(B) 31.2
(C) 47.1
(D) 50.1

Q12. If a and b are the roots of $4x^2-3x+7=0$ then the value of $1/a+1/b$ is .....

(A) $-3/7$
(B) $3/7$
(C) $-3/4$
(D) $4/7$

Q13. If $m \propto 1/n$ then .....

(A) $m=kn$
(B) $n=km$
(C) $m/n=k$
(D) $mn=k$

Q14. $(x^2+2)/(x^2+2x+2)$ is ..... fraction.

(A) proper
(B) improper
(C) irrational
(D) none of these

Q15. If f is a function from A to B then f is onto function if .....

(A) Range $f \neq B$
(B) Dom $f=A$
(C) Range $f=B$
(D) none of these

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 36

Q1. Attempt any nine of the following short questions. Each question carries 4 marks.

(i) Solve the equation $6x^4-13x^2+5=0$

(ii) For what value of k, the roots of the equation $kx^2+4x+1=0$ are real?

(iii) Show that $(-1+i\sqrt{3})^4(-1-i\sqrt{3})^5 = 512w^2$

(iv) If -1 and 2 are of the quadratic equation $x^4-5x^2+4=0$. Use synthetic division to find the other roots.

(v) If x varies inversely to y and x=3 when y=12. Find the value of y when x=6.

(vi) Resolve into partial fraction: $(x-1)/(x^2+6x+5)$

(vii) If U={1,2,3,4,5,6}, A={2,3} and B={3,4,5}, find A'∩B' and A'∪B'

(viii) A test was conducted in a class of 13 students and the following marks were obtained by the students: 10,12,12,14,9,18,9,13,16,9,17,18,14. Make frequency table and find arithmetic mean.

(ix) Prove that $\tan\theta+\cot\theta = \sec\theta\operatorname{cosec}\theta$

(x) Solve the system of equations: $2x-y=-8$, $x^2+4x=y$

(xi) The angle of elevation of the top of a cliff is 30°, walking 210 meter from the point towards the cliff, the angle of elevation becomes 45°. Find the height of the cliff.

(xii) Find variance and standard deviation for the following data: 6,8,10,12,14

SECTION-C

Maximum Marks: 24

> Note: Attempt any three of the following questions. Each question carries 8 marks.

Q2. Prove that if two chords of a circle are congruent then they will be equidistant from the centre.

Q3. Show that the two tangents drawn to a circle from a point outside it, are equal in length.

Q4. Prove that any two angles in the same segment of a circle are equal.

Q5. Radii of two circles are 2cm and 3cm and their centres are 8cm apart. Draw direct common tangents to the circles.